

Chapter 3

GENERALIZED HOOKE'S LAW

This chapter is devoted to the relation between the stress and strain tensors. As well as the classical linear Hooke's law, which characterizes a wide class of elastic materials such as e.g. steel, concrete, glass, ceramic material, wood, etc., and the interpretation of this law, we present also a special case of a nonlinear Hooke's law leading to the so-called deformation theory of plasticity. Hencky material is also discussed as a limit of the previous case.

The deformation theory can be applied only to special problems. The flow theory presented in this chapter is of a general character. This theory gives a reliable basis for a generally valid theory of material subjected to both elastic and plastic deformation. The flow theory is a limiting case of the theory of elasto-inelastic material with internal state variables. The constitutive equations for such material are also given in Chapter 3.

3.1. TENSION TEST

Consider a bar of length l and (constant) cross-section q . We take the x_1 -axis in the direction of the axis of the bar and subject the bar to the tensile force P at both its

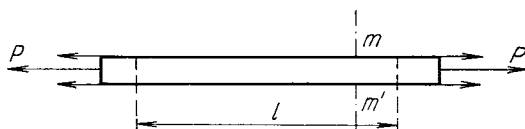


Fig. 9

ends (Fig. 9). Suppose that the normal stress τ_{11} is constant throughout the entire bar. The equilibrium of that part of the bar that is to the left of the plane section $m - m'$ then implies

$$\tau \equiv \tau_{11} = \frac{P}{q}$$

if we neglect the fact that the area of the cross-section decreases owing to the extension of the bar. (Considering small strain, we shall see in what follows that this decrease

is a higher order quantity.) Measuring now the relative extension $e = \Delta l/l$ as a function of the stress τ , we obtain the so-called *tension test diagram* (see Fig. 10, which corresponds to steel).

The part of the curve between the points O and A is straight. The corresponding stress σ_A is called the *proportional limit*. The part $\widehat{AB}$ is curved and the extension increases suddenly at the point B without any change in the stress. The stress σ_B is called

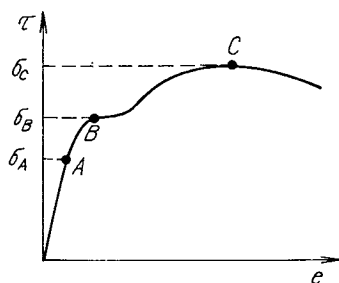


Fig. 10

the *yield point*. If the stress further increases, the extension increases as well up to the point C . The stress σ_C corresponding to this point C is called the *strength limit*. Further stretching would finally cause the bar to break.

Let us stop the increase of the force P at a certain point of the curve and now make the force decrease to zero. If this reversal takes place at a sufficiently small stress τ we shall come back to the origin along the same curve. The maximum of such stresses is called the *elastic limit* σ_E . As a rule,

$$\sigma_E \geq \sigma_A.$$

The deformation corresponding to $\tau \leq \sigma_E$ is called *elastic*. In other words, the elastic deformation (extension) vanishes completely if the load is entirely removed.

However, if the load was removed after the stress had exceeded the value σ_E a part of the extension vanishes, although part of it remains permanently. This part of the deformation is called *plastic*. The magnitude of the permanent deformation depends on the nature of material, temperature, etc.

Elastic deformations are studied in the *theory of elasticity* while the *theory of plasticity* deals with plastic deformations.

For certain materials, the straight part of the diagram is so short that it is practically negligible, but the deformation can be regarded as elastic even for relatively large stresses (i.e., $\sigma_E > \sigma_A \doteq 0$).

Similar results are obtained in the compression, bending or torsion tests.

In what follows we shall mainly be concerned with the *linear theory of elasticity* that corresponds to the stresses $\tau < \sigma_A$. Then e and τ are mutually proportional, which is expressed by *Hooke's law*

$$\tau = Ee,$$

where E is the so-called *Young's modulus of elasticity*.

3.2. GENERALIZED HOOKE'S LAW

In Section 3.1 we studied the “uniaxial” case only, i.e., the strain in the direction of the acting stress τ_{11} . We shall now extend Hooke's law to the general case of spatial (triaxial) strain and stress.

Suppose that at the point $\mathbf{x} \in \Omega$ there exists a linear relation between the stress tensor and the small strain tensor, i.e.,

$$\tau_{ij} = c_{ijkl} e_{kl} \quad (i, j = 1, 2, 3). \quad (2.1)$$

There is no constant term in this relation since we suppose that the stress tensor vanishes whenever the strain tensor is equal to zero.

The conditions

$$c_{ijkl} = c_{jikl}, \quad c_{ijkl} = c_{ijlk} \quad (2.2)$$

follow from the symmetry of the tensors τ_{ij} and e_{ij} . We further obtain the relations

$$c_{ijkl} = c_{klij} \quad (2.3)$$

from energy considerations. Thus there are at most 21 independent constants c_{ijkl} that together characterize general material at the point $\mathbf{x}$. Since e.g. blue-stone indeed has 21 independent constants, this number cannot be reduced in general.

Definition 2.1. If the constants $c_{ijkl}(\mathbf{x})$ do not depend on $\mathbf{x} \in \Omega$, the material of the body is said to be *homogeneous*.

If the constants $c_{ijkl}(\mathbf{x})$ do not depend on the choice of coordinate system, the material of the body is said to be *isotropic* at the point $\mathbf{x}$. Otherwise, the material is *anisotropic* at the point $\mathbf{x}$.

Remark 2.1. For example, wood is anisotropic at any point since it possesses different properties along and across its fibres. (Such material is called “orthotropic”.)

Theorem 2.1. The generalized Hooke's law for isotropic material states that

$$\tau_{ij}(\mathbf{x}) = \lambda(\mathbf{x}) \delta_{ij} \vartheta(\mathbf{x}) + 2 \mu(\mathbf{x}) e_{ij}(\mathbf{x}), \quad (2.4)$$

where

$$\mathfrak{I}(\mathbf{x}) = e_{ii}(\mathbf{x})$$

and λ and μ are the so-called *Lamé coefficients*.

Remark 2.2. $\lambda = c_{1122}$, $\mu = c_{1212}$. We shall omit subscript 1 of the first invariant $\mathfrak{I}_1$.

Proof of the theorem will be accomplished in three steps.

1. In isotropic material, the principal directions of (small) strain coincide with the principal directions of stress. Proof: We can take the coordinate axes in the principal directions of strain. Then $e_{12} = e_{13} = e_{23} = 0$. We shall show that $\tau_{23} = 0$. We first have

$$\tau_{23} = Ae_{11} + Be_{22} + Ce_{33}.$$

We now rotate the coordinate system through an angle of 180° about the x_3 -axis. Then $x'_1 = -x_1$, $x'_2 = -x_2$, and $x'_3 = x_3$ and the matrix of the transformation is

$$\mathbf{a} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

Therefore

$$\tau'_{23} = a_{2i}a_{3j}\tau_{ij} = -\tau_{23},$$

$$e'_{11} = a_{1i}a_{1j}e_{ij} = e_{11},$$

$$e'_{22} = a_{2i}a_{2j}e_{ij} = e_{22},$$

$$e'_{33} = e_{33}.$$

The relation

$$\tau'_{23} = Ae'_{11} + Be'_{22} + Ce'_{33} = Ae_{11} + Be_{22} + Ce_{33} = \tau_{23}$$

is now a consequence of isotropy since the constants A , B , and C do not depend on the coordinate system. Thus

$$-\tau_{23} = \tau'_{23} = \tau_{23} \Rightarrow \tau_{23} = 0.$$

We could similarly show that $\tau_{12} = \tau_{13} = 0$.

2. Consider the component τ_{11} . Taking the coordinate axes in the principal directions of strain, we obtain

$$\tau_{11} = ae_{11} + be_{22} + ce_{33}. \quad (\text{A})$$

We now rotate the coordinate system through an angle of 90° about the x_1 -axis in such a way that

$$x'_1 = x_1, \quad x'_2 = x_3, \quad x'_3 = -x_2.$$

The matrix of this transformation is

$$\mathbf{a} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}.$$

We have

$$\tau'_{11} = a_{1i}a_{1j}\tau_{ij} = \tau_{11},$$

$$e'_{11} = e_{11}, \quad e'_{22} = a_{2i}a_{2j}e_{ij} = e_{33}, \quad e'_{33} = e_{22}$$

and therefore

$$\tau'_{11} = ae'_{11} + be'_{22} + ce'_{33} = ae_{11} + be_{33} + ce_{22}.$$

This implies $b = c$ since $\tau'_{11} = \tau_{11} = ae_{11} + be_{22} + ce_{33}$. We can thus write τ_{11} as

$$\tau_{11} = ae_{11} + b(e_{22} + e_{33}) = (a - b)e_{11} + b\vartheta = \lambda\vartheta + 2\mu e_{11}.$$

The same relation can also be obtained for subscripts 2 and 3. Altogether we have

$$\tau_{11} = \lambda\vartheta + 2\mu e_{11},$$

$$\tau_{22} = \lambda\vartheta + 2\mu e_{22},$$

$$\tau_{33} = \lambda\vartheta + 2\mu e_{33},$$

$$\tau_{ij} = 0 \quad \text{for } i \neq j,$$

which is the generalized Hooke's law for isotropic material in the principal directions.

3. The tensors τ_{ij} and e_{ij} of order two transform according to (1.1.2), namely as a quadratic form. The quadratic form in the principal directions is

$$\begin{aligned} & \tau_{11}\xi_1^2 + \tau_{22}\xi_2^2 + \tau_{33}\xi_3^2 = \\ & = \lambda\vartheta(\xi_1^2 + \xi_2^2 + \xi_3^2) + 2\mu(e_{11}\xi_1^2 + e_{22}\xi_2^2 + e_{33}\xi_3^2). \end{aligned}$$

Passing to the new coordinate system, we obtain

$$\tau'_{ij}\xi'_i\xi'_j = \lambda\vartheta(\xi_1'^2 + \xi_2'^2 + \xi_3'^2) + 2\mu e'_{ij}\xi'_i\xi'_j.$$

We thus have two quadratic forms which are identical. This implies that their corresponding coefficients are equal. Therefore

$$\tau'_{ij} = \lambda\vartheta\delta_{ij} + 2\mu e'_{ij}. \quad \square$$

We shall establish certain inequalities for the Lamé coefficients in what follows.

Suppose that *there exists a law that is inverse* to the generalized Hooke's law, i.e., that we can calculate the components of the small strain tensor e_{kl} from the stress tensor τ_{ij} . The system (2.1) can thus be solved for e_{ij} uniquely. The determinant of the 6×6 matrix of the system (2.4) is equal to $(2\mu)^5 (3\lambda + 2\mu)$. Therefore, it follows from our assumption that

$$\mu \neq 0, \quad 3\lambda + 2\mu \neq 0.$$

We shall analyze several hypothetical experiments under the following assumptions: (a) for the stress tensor, we substitute functions satisfying the equations of equilibrium (1.4.2); (b) for the strain tensor, we substitute functions satisfying the equations of compatibility (2.3.2); (c) the generalized Hooke's law (2.4) is assumed to hold for the stress and strain tensors.

Example 2.1. A homogeneous cube is subjected to hydrostatic pressure alone. Suppose that the size of the cube may be neglected in comparison with depth. We then put

$$T_i = \tau_{ij}v_j = -pv_i, \quad p = \text{const} > 0 \quad (i = 1, 2, 3).$$

We shall verify that the stress tensor $\tau_{ij} = -p\delta_{ij}$ satisfies the conditions (a), (b), and (c). The equations of equilibrium and compatibility are clearly satisfied since $\tau_{ij} = \text{const}$ and $e_{ij} = \text{const}$. (λ and μ are constant in the whole cube.) The studied tensor satisfies the boundary conditions, too, since

$$T_i = -p\delta_{ij}v_j = -pv_i.$$

In Section 7.4, we shall show that our problem has a unique solution expressed in terms of stresses. The tensor $\tau_{ij} = -p\delta_{ij}$ thus represents the "real" stress.

Consider the first invariants of the stress and strain tensors. Obviously $\theta = \tau_{ii} = -3p$. Summing the first three equations of (2.4) we obtain

$$\theta = (3\lambda + 2\mu) \vartheta, \tag{2.5}$$

i.e.,

$$-3p = (3\lambda + 2\mu) \vartheta. \tag{2.6}$$

We know from (2.2.3) that the meaning of the invariant ϑ is the cubical dilatation,

$$\vartheta \doteq \frac{V' - V}{V}.$$

In the case of hydrostatic pressure, clearly $V' < V$ and thus $\vartheta < 0$. From (2.6), we now conclude that $3\lambda + 2\mu > 0$. The expression

$$k = \lambda + \frac{2}{3}\mu \tag{2.7}$$

is called the *bulk modulus*.

Example 2.2. Consider a cube subjected only to a constant shear $T > 0$ on the four faces:

$$x_1 = h : T_i = T\delta_{2i},$$

$$x_1 = 0 : T_i = -T\delta_{2i},$$

$$x_2 = h : T_i = T\delta_{1i},$$

$$x_2 = 0 : T_i = -T\delta_{1i}.$$

For $x_3 = 0$ and $x_3 = h$, we put $T_i = 0$. We shall show that the tensor

$$(\tau_{ij}) = \begin{bmatrix} 0 & T & 0 \\ T & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

i.e.,

$$\tau_{ij} = T(\delta_{i1}\delta_{j2} + \delta_{i2}\delta_{j1}),$$

is the solution in terms of stresses.

The equations of equilibrium and compatibility are clearly satisfied. The boundary conditions, e.g. for $x_1 = h$, are

$$v_j = \delta_{1j},$$

$$\tau_{ij}v_j = T(\delta_{i1}\delta_{j2} + \delta_{i2}\delta_{j1})\delta_{1j} = T\delta_{i2},$$

etc. In view of (2.5), the cubical dilatation is

$$\vartheta = \frac{\theta}{3\lambda + 2\mu} = 0$$

since $\theta = 0$. Therefore

$$e_{ij} = \frac{1}{2\mu} \tau_{ij}$$

follows from (2.4). We know that $2e_{12}$ represents the angle of shear (see (2.2.2) and Fig. 7), i.e., the reduction of the right angle between the directions of the x_1 - and x_2 -axes. In our case it is clear (see Fig. 11) that this angle decreases and we thus have $2e_{12} \doteq \lim_{t \rightarrow 0+} \alpha_{12}(t) > 0$. From the relation

$$2e_{12} = \frac{1}{\mu} \tau_{12} = \frac{1}{\mu} T > 0$$

we then conclude that

$$\mu > 0. \quad (2.8)$$

Example 2.3. Consider a circular cylinder, take the x_1 -axis in the direction of the axis of the cylinder, and subject the cylinder to a tension $T = \text{const} > 0$ at its ends, i.e.,

$$x_1 = l : T_i = T\delta_{i1},$$

$$x_1 = 0 : T_i = -T\delta_{i1}$$

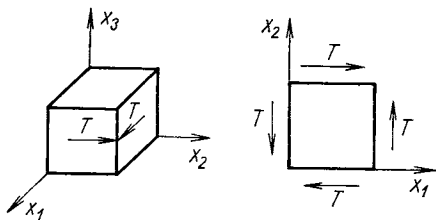


Fig. 11

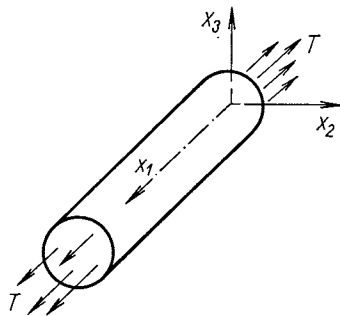


Fig. 12

(see Fig. 12). No load acts on the curved surface of the cylinder. We therefore have $T_i = 0$ for $x_2^2 + x_3^2 = r^2$. We shall show that

$$(\tau_{ij}) = \begin{bmatrix} T & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

i.e., $\tau_{ij} = T\delta_{i1}\delta_{j1}$, is the solution expressed in terms of stresses. Indeed, the equations of equilibrium and compatibility are satisfied and it remains to check the boundary conditions. We have e.g. $v_j = \delta_{1j}$ and

$$\tau_{ij}v_j = T\delta_{i1}\delta_{j1}\delta_{j1} = T\delta_{i1}$$

on the face $x_1 = l$. Taking into account that $\mathbf{v} = (0, x_2/r, x_3/r)$ on the curved surface, we obtain

$$\tau_{ij}v_j = \tau_{i2}\frac{x_2}{r} + \tau_{i3}\frac{x_3}{r} = 0.$$

The relations (2.4) and (2.5) imply

$$\vartheta = \frac{\theta}{3\lambda + 2\mu},$$

$$e_{ij} = \frac{1}{2\mu} \left(\tau_{ij} - \frac{\lambda}{3\lambda + 2\mu} \theta \delta_{ij} \right), \quad (2.9)$$

which gives

$$e_{11} = \frac{1}{2\mu} \left(T - \frac{\lambda}{3\lambda + 2\mu} T \right) = \frac{\lambda + \mu}{\mu(3\lambda + 2\mu)} T. \quad (2.10)$$

In Sec. 3.1 we established Hooke's law for a "uniaxial" tension in the form

$$\frac{\Delta l}{l} = \frac{1}{E} \tau = \tau_{11}.$$

In our case $e_{11} = \partial u_1 / \partial x_1 = \text{const}$ and finally

$$\Delta l = u|_{x=l} - u|_{x=0} = \int_0^l e_{11} dx_1 = l e_{11}.$$

Comparing this with the formula (2.10) we obtain

$$\frac{1}{E} = \frac{\lambda + \mu}{\mu(3\lambda + 2\mu)}. \quad (2.11)$$

Further

$$e_{22} = e_{33} = - \frac{\lambda}{2\mu(3\lambda + 2\mu)} T.$$

For $T > 0$, transverse contraction of the cylindrical rod takes place, i.e., $e_{22} < 0$ and $e_{33} < 0$. Employing (2.7) and (2.8), we conclude from this that

$$\lambda > 0. \quad \square \quad (2.12)$$

We write

$$\sigma = \left| \frac{e_{22}}{e_{11}} \right| = \frac{\lambda}{2(\lambda + \mu)}. \quad (2.13)$$

This number is called *Poisson's ratio*.

In engineering, Young's modulus E of elasticity and Poisson's ratio σ are used more often than the Lamé coefficients λ and μ . Finally we express the Lamé coefficients from (2.11) and (2.13) in terms of E and σ . From (2.13) we obtain that $0 < \sigma < \frac{1}{2}$ in view of (2.12) and (2.8). Further

$$\lambda = \frac{2\sigma\mu}{1 - 2\sigma}.$$

Then

$$\frac{\sigma}{E} = \frac{\lambda}{2\mu(3\lambda + 2\mu)} = \frac{2\sigma\mu}{1 - 2\sigma} \frac{1}{2\mu(3(2\sigma\mu)/(1 - 2\sigma) + 2\mu)} = \frac{\sigma}{2\mu(1 + \sigma)},$$

from which

$$\mu = \frac{E}{2(1 + \sigma)}, \quad (2.14)$$

$$\lambda = \frac{E\sigma}{(1 + \sigma)(1 - 2\sigma)}. \quad (2.15)$$

We clearly have that

$$\lambda > 0, \quad \mu > 0 \Leftrightarrow E > 0, \quad 0 < \sigma < \frac{1}{2}.$$

Several values of the modulus of elasticity and Poisson's ratio are given in the following table.

	$E \text{ (Nm}^{-2}\text{)}$	σ
aluminium	0.73×10^{11}	0.34
copper	1.18×10^{11}	0.35
iron	2.13×10^{11}	0.28
steel	2.15×10^{11}	0.29

3.3. ELASTO-PLASTIC MATERIALS. DEFORMATION THEORY. (A SPECIAL CASE OF THE NONLINEAR HOOKE'S LAW)

We shall suppose that the Lamé coefficients depend on the first two invariants ϑ_1 and ϑ_2 of the small strain tensor and, moreover, that the relation between the first invariants of stress and strain is linear. This implies that

$$\tau_{ij} = [k(\mathbf{x}) - \frac{2}{3}\mu(\mathbf{x}, \Gamma)] \vartheta_1 \delta_{ij} + 2\mu(\mathbf{x}, \Gamma) e_{ij}, \quad (3.1)$$

where $k(\mathbf{x})$ is the bulk modulus (cf. (2.7)), and $\Gamma = \sqrt{\gamma}$, where

$$\gamma \equiv \frac{1}{3}[(e_{11} - e_{22})^2 + (e_{22} - e_{33})^2 + (e_{11} - e_{33})^2 + 6(e_{12}^2 + e_{23}^2 + e_{13}^2)].$$

We readily find that

$$\gamma = -\frac{1}{3}\vartheta_1^2 + e_{ij}e_{ij} = \frac{2}{9}(\vartheta_1^2 - 3\vartheta_2). \quad (3.2)$$

Introducing the deviators of the stress and strain tensors in the usual way, i.e., by the relation

$$\tau'_{ij} \equiv \tau_{ij} - \tau \delta_{ij}, \quad e'_{ij} = e_{ij} - e \delta_{ij}, \quad (3.3)$$

where $\tau = \frac{1}{3}\tau'_{ii}$ and $e = \frac{1}{3}e'_{ii}$, we then obtain

$$\tau = 3ke, \quad (3.4)$$

$$\tau'_{ij} = 2\mu e'_{ij}, \quad (3.5)$$

$$\gamma = e'_{ij}e'_{ij} \quad (3.6)$$

from (3.1) (cf. also Section 2.3 where $2\varepsilon^2 \equiv \gamma = \Gamma^2$).

For the stress tensor we can similarly introduce the invariant

$$s = \tau'_{ij}\tau'_{ij}, \quad (3.7)$$

which represents the intensity of shear stress (see Definition 1.6.3, where $2S^2 \equiv s$). Returning to (3.5) and putting $S \equiv \sqrt{s}$, we obtain

$$S = 2\mu\Gamma, \quad (3.8)$$

which is the so-called Hencky relation.

It is natural to suppose that

$$0 < \mu_0 \leq \mu(\mathbf{x}, \Gamma) \leq \mu_1 < \infty \quad (3.9)$$

and that $dS/d\Gamma > 0$; from this we have that $dS/d\Gamma = 2(\mu + \mu'\Gamma) > 0$.

We shall thus suppose that

$$\mu + \mu'\Gamma \geq \alpha > 0. \quad (3.10)$$

Putting

$$A \equiv \frac{k}{2} g_1^2 + 2 \int_0^\Gamma \mu(t) t dt = \frac{k}{2} g_1^2 + \int_0^\Gamma S(t) dt, \quad (3.11)$$

we easily find that

$$\tau_{ij} = \frac{\partial A}{\partial e_{ij}}. \quad (3.12)$$

The relation (3.12) is a general nonlinear Hooke's law in the small displacement theory. We suppose that the function $A(e_{ij})$ does not depend on the substitution of e_{ji} for e_{ij} . Elasto-plastic material is thus governed by a law that is a special case of a nonlinear Hooke's law.

3.4. ELASTO-INELASTIC BODIES. A MODEL WITH INTERNAL STATE VARIABLES

Suppose that the small strain tensor can be split into two parts, elastic and plastic, i.e.,

$$e_{ij} = e_{ij}^e + e_{ij}^p. \quad (4.1)$$

(In the foregoing sections we had $e_{ij}^p \equiv 0$.)

We further suppose that both e_{ij}^e and e_{ij}^p are symmetric tensors. Moreover, let the relation

$$e_{ij}^e = \frac{\partial P}{\partial \tau_{ij}}(\tau, \alpha) \quad (4.2)$$

hold between the tensor e_{ij}^e and the stress tensor τ_{ij} . This is — in general — a nonlinear relation also depending on the *internal state variables* $(\alpha_1, \alpha_2, \dots, \alpha_n)$. The function P in (4.2) is assumed to have Lipschitz-continuous derivatives and to be invariant with respect to the substitution of τ_{ji} for τ_{ij} . We also suppose that

$$\frac{\partial^2 P}{\partial \tau_{ij} \partial \tau_{kl}}(\tau, \alpha) \eta_{ij} \eta_{kl} \geq C_1 \eta_{ij} \eta_{ij} \quad (4.3)$$

for any $\eta_{ij} = \eta_{ji}$ and any α , which is the usual condition of positive definiteness.

Formulating boundary value problems, we consider a quasi-static process in this case. We therefore assume that the boundary conditions as well as the studied displacements, strain and stress tensors, and internal state variables are functions of a variable t (e.g., of time). Consequently, the relation (4.2) depends on a parameter and the resulting solution depends on “history”.

In addition to the law (4.2), we shall suppose that additional constitutive equations hold:

$$\frac{de_{ij}^p}{dt} = B_{ij}(\tau, \alpha), \quad (4.4)$$

$$\frac{d\alpha_i}{dt} = C_i(\tau, \alpha). \quad (4.5)$$

Moreover, if the process starts at the point $t = 0$ we assume that $e_{ij}^p(0)$ and $\alpha_i(0)$ are known quantities. The functions B_{ij} and C_i are assumed to be uniformly Lipschitz-continuous.

The detailed formulation of boundary value problems for an elasto-inelastic body and some existence theorems can be found in Chapter 9. The model with internal state variables may be treated as a certain regularization of the very popular flow model presented e.g. in the book by Washizu (1968). The flow model is, in a sense, a limit of models with internal state variables (see the paper by Kratochvíl and Trávníček (1977), and Section 13.5).

3.5. HOOKE'S LAW WITH A PERFECTLY PLASTIC DOMAIN

This section is closely related to Section 3.3. First we shall consider a picture of the function $S = S(\Gamma) = 2\mu(\Gamma) \Gamma$ defined in Sec. 3.3 (see Fig. 13). The function $\Gamma \mapsto S(\Gamma)$

is increasing since $dS/d\Gamma > 0$ by hypothesis. We can thus define the function $v(\mathbf{x}, S)$ by the formula

$$\Gamma = 2vS, \quad (5.1)$$

where $v(\mathbf{x}, S) > 0$ for $S > 0$. From this we have

$$e'_{ij} = 2v\tau'_{ij}. \quad (5.2)$$

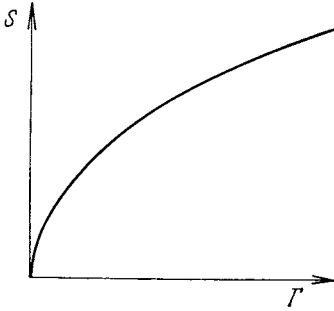


Fig. 13

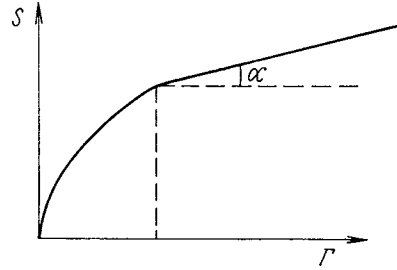


Fig. 14

Let the graph of the function $S = S(\Gamma)$ be of the form shown in Fig. 14 where $\alpha > 0$. Passing to the limit as $\alpha \rightarrow 0$, we obtain

$$S = 2\mu(\Gamma)\Gamma \quad \text{for } \Gamma < \Gamma_0, \quad (5.3)$$

$$S = 2\mu(\Gamma_0)\Gamma_0 \quad \text{for } \Gamma \geq \Gamma_0 \quad (5.4)$$

and put $S_0 = 2\mu(\Gamma_0)\Gamma_0$. For $\Gamma \geq \Gamma_0$, (5.4) gives

$$\mu(\Gamma) = \mu(\Gamma_0) \frac{\Gamma_0}{\Gamma}. \quad (5.5)$$

In this case we thus have

$$\tau'_{ij} = 2\mu(\Gamma)e'_{ij} \quad \text{for } \Gamma \leq \Gamma_0, \quad (5.6)$$

$$\tau'_{ij} = 2\mu(\Gamma_0) \frac{\Gamma_0}{\Gamma} e'_{ij} \quad \text{for } \Gamma > \Gamma_0. \quad (5.7)$$

The relation (3.4) remains unchanged.

The material that is governed by the law (5.6) and (5.7) is called *perfectly plastic* in the domain $\Gamma > \Gamma_0$. In other words, the material is elasto-plastic for $S < S_0$ and perfectly plastic for $S = S_0$. The condition $S = S_0$ is called the *Mises yield condition*.

Studying the converse of the relation (5.6), i.e., the equality (5.2) as $\alpha \rightarrow 0$, and supposing that the stress and strain tensors are also defined after passing to the limit, we obtain $S = 2\mu\Gamma = 2\mu \, 2\nu S$ for $\alpha > 0$ from (3.8) and (5.1). Finally

$$\nu = \frac{1}{4\mu}. \quad (5.8)$$

If $\Gamma > \Gamma_0$ after passing to the limit, then

$$\lim_{\alpha \rightarrow 0} \nu(S) = \frac{1}{4\mu(\Gamma_0)} \frac{\Gamma}{\Gamma_0} > \nu(S_0).$$

This implies that

$$e'_{ij} = 2\nu(S_0) \tau'_{ij} + \lambda \tau'_{ij}, \quad S = S_0, \quad (5.9)$$

for the perfectly plastic domain, where $\lambda > 0$ is an undetermined function of $\mathbf{x}$.

As a matter of course, the relation (5.2) remains true and (3.4) also shows no change for $S < S_0$. Material governed by the relations (3.4), (5.2), and (5.9) or by the relations (3.4), (5.6), and (5.7) is called *Hencky material*.

3.6. FLOW THEORY OF PLASTICITY

As in Section 3.4, we again assume that the small strain tensor can be split into its elastic and plastic parts

$$e_{ij} = e_{ij}^e + e_{ij}^p, \quad (6.1)$$

where e_{ij}^e and e_{ij}^p are symmetric tensors. We suppose that the relation between the tensor e_{ij}^e and the stress tensor τ_{ij} is expressed by the linear Hooke's law

$$e_{ij}^e = A_{ijkl} \tau_{kl}. \quad (6.2)$$

Just as in Section 3.4, we consider a quasi-static process. The studied displacements, and strain and stress tensors are therefore functions of a variable t (e.g., of time).

Moreover, let a function $f(\boldsymbol{\tau})$, the so-called yield condition, and a positive constant $\bar{k}$ be given. The surfaces $f(\boldsymbol{\tau}) = c$ where $c \geq \bar{k}$ are the so-called yield surfaces. In addition, let a positive function $h(\boldsymbol{\tau})$ be given. Consider a point $\mathbf{x}$ of the studied body Ω and a parameter t . We further suppose that

$$n(\boldsymbol{\tau})(\mathbf{x}, t) = \max(\bar{k}, \max_{0 \leq s \leq t} f(\boldsymbol{\tau}(\mathbf{x}, s))).$$

If now $f(\boldsymbol{\tau}(\mathbf{x}, t)) < n(\boldsymbol{\tau})(\mathbf{x}, t)$, then the increment of the plastic strain is zero. If $f(\boldsymbol{\tau}(\mathbf{x}, t)) = n(\boldsymbol{\tau})(\mathbf{x}, t)$ holds we put

$$e_{ij}^p = h \frac{d}{dt}(f) \frac{\partial f}{\partial \tau_{ij}}; \quad (6.3)$$

both cases can be expressed by

$$\dot{e}_{ij}^p = h \frac{d}{dt}(n) \frac{\partial f}{\partial \tau_{ij}}. \quad (6.4)$$

The relations (6.2) and (6.4) define the flow theory with strain hardening. We shall discuss this theory in detail in Sec. 13.5; see also Washizu (1968). The hardening is caused by the fact that the yield surfaces are moved by stress on a higher level that is proportional to plastic work (cf. Washizu, 1968). However, we can also consider the case when the yield surface is fixed and is described by the equation $f(\boldsymbol{\tau}) = \bar{k}$. In this case, we again have $\dot{e}_{ij}^p = 0$ if $f(\boldsymbol{\tau}(\mathbf{x}, t)) < \bar{k}$ and $\dot{e}_{ij}^p \geq 0$ if the converse is true. We shall study this subject in more detail in Sec. 13.4 (see also e.g. Washizu, 1968).